

federated operator crates
(each: own crate + version + commit_hash)

apsis-1pn
1PN — Hamiltonian
Anderson 1975

apsis-central
central force — Ham.
Tamayo 2020

apsis-radiation
PR-drag — non-cons.
Burns 1979

each publishes:
declared_units ·
kernel_requirements ·
check_regime · citation

registration gate `System::add*_perturbation`
HARD unit mismatch → `Err(UnitSystemMismatch)`
— *not registered*, caller decides
SOFT kernel / regime violation →
`warn_diag` — registers, integration proceeds

apsis core engine
IAS15 integrator loop · Newton kernel
· `System` registry · structured log bus

apsis-python
apsis-py-core
`Err` → typed exception

trajectory + provenance (`cite` → BibTeX +
`Cargo.lock` hash) + audit log (`warn_diag` events)